

Nutritional practices on Manitoba dairy farms

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Plaizier, J. C., Garner, T., Droppo, T. and Whiting, T. 2004. **Nutritional practices on Manitoba dairy farms.** *Can. J. Anim. Sci.* **84**: 501–509. A survey was conducted on 40 randomly selected dairy farms across Manitoba to document nutritional practices and diet compositions, and study relationships between diet composition and milk production. Samples from all feeds, diets, and the bulk milk tank were collected and analyzed. Inclusion rates of feed ingredients were recorded. Production data were obtained from Western Canada Dairy Herd Improvement Services (WCDHIS). Component feeding and total mixed ration (TMR) feeding were used in 37.5 and 62.5% of herds, respectively. Only 24% of TMR-fed herds used two or more TMR. The medians of dietary contents of crude protein (CP), rumen degradable protein (RDP), rumen undegradable protein (RUP), and neutral detergent fibre (NDF), and milk urea nitrogen (MUN) were 18.3% dry matter (DM), 12.2% DM, 5.7% DM, 34.8% DM, and 15.6 mg dL⁻¹, respectively. On average, diets contained 6.2% more net energy for lactation (NEI), 35.9% more RDP, 15.8% less RUP, 74.6% more calcium (Ca), 51.9% more phosphorous (P), 44.9% more potassium (K), 116.1% more magnesium (Mg), and 4.2% more sodium (Na) than the estimated requirements based on the average milk production and estimated DM intake of the cows on each farm. In 25% of TMR-fed herds, TMR was coarser than recommended. Milk yield, and milk fat percentage were affected by breed, but were not affected by feeding practice (TMR or component feeding), diet composition, and physically effective NDF (peNDF) determined as the percentage of feed particles retained by the 8- and 19-mm screens of the Penn State Particle Separator multiplied by dietary NDF. Milk protein percentage was positively correlated to RUP. MUN was positively correlated to RDP, RUP, NDF, and days in milk (DIM). Reductions in dietary CP, RDP, Ca, P, Mg, and K could reduce nutrient excretions to the environment without reducing milk production and health. Increasing dietary RUP content could improve milk production on Manitoba dairy farms.

Key words: Dairy cattle, milk production, protein, fibre, minerals, particle size.

Plaizier, J. C., Garner, T., Droppo, T. et Whiting, T. 2004. **Pratiques nutritionnelles dans les élevages de bovins laitiers du Manitoba.** *Can. J. Anim. Sci.* **84**: 501–509. Les auteurs ont effectué une enquête dans 40 exploitations laitières du Manitoba sélectionnées au hasard. L'objectif était d'étayer les pratiques alimentaires et la composition des rations, ainsi que d'étudier les liens entre cette dernière et la production laitière. On a prélevé puis analysé des échantillons de tous les aliments du bétail, des rations et de lait gardé en vrac dans les citernes de stockage. On a aussi noté les taux d'incorporation des ingrédients de la ration. Les données sur la production viennent des Western Canada Dairy Herd Improvement Services (WCDHIS). On sert des ingrédients ou un mélange complet dans 37,5 % et 62,5 % des troupeaux, respectivement. Vingt-quatre pour cent des bêtes nourries avec un mélange seulement reçoivent deux mélanges ou plus. La médiane pour les aliments s'établit comme suit : 18,3 % de la matière sèche (MS) pour les protéines, 12,2 % de la MS pour les protéines dégradables dans le rumen (PDR), 5,7 % de la MS pour les protéines non dégradables (PNR), 34,8 % de la MS pour les fibres au détergent neutre (FDN) et 15,6 mg d'azote uréique par décilitre de lait. En moyenne, la ration renferme 6,2 % de plus d'énergie nette pour la lactation, 35,9 % de plus de PDR, 15,8 % de plus de PNR, 74,6 % plus de calcium (Ca), 51,9 % plus de phosphore (P), 44,9 % plus de potassium (K), 116,1 % plus de magnésium (Mg) et 4,2 % plus de sodium (Na) que ce que les animaux devraient avoir besoin compte tenu de la production moyenne de lait et de la quantité de matière sèche estimative ingérée dans chaque exploitation. Pour 25 % des exploitations où les animaux reçoivent un mélange, ce dernier a une granulométrie plus grossière que celle recommandée. Le rendement laitier et le pourcentage de matière grasse butyrique sont affectés par la race mais pas par les pratiques alimentaires (mélange ou ingrédients), la composition de la ration et les FDN efficaces physiquement (pourcentage de particules retenues par les tamis à mailles de 8 et de 19 mm du séparateur Penn State multiplié par la proportion de FDN dans la ration). Le pourcentage de protéines dans le lait présente une corrélation positive avec les PDR, les PNR, les FDN et le nombre de jours en lactation. Une diminution de la concentration de PDR, de Ca, de P, de Mg et de K dans l'aliment pourrait réduire l'excrétion d'oligo-éléments dans l'environnement sans nuire à la production de lait ni à la santé des animaux. Une hausse de la concentration de PNR dans la ration pourrait accroître la production de lait dans les exploitations du Manitoba.

Mots clés: Bovins laitiers, production laitière, protéines, fibres, oligo-éléments, granulométrie

Milk yields of Canadian dairy cows continue to steadily increase. The prediction of nutrient requirements and the formulation of diets need to reflect the rapid changes of

dairy cattle productivity. As a result, the 2001 Nutrient Requirements of Dairy Cattle [National Research Council (NRC) 2001] contains many changes over the previous ver-

Abbreviations: CP, crude protein; DIM, days in milk; DM, dry matter; DMI, dry matter intake; MP, metabolizable protein; MUN, milk urea nitrogen; NDF, neutral detergent fibre; NEI, net energy for lactation; NFC, non-fibre carbohydrates; peNDF, physically effective NDF; PSPS, Penn State Particle Separator; RDP, rumen degradable protein; RUP, rumen undegradable protein; TMR, total mixed ration, WCDHIS, Western Canada Dairy Herd Improvement Services

sion of these requirements (NRC 1989). It is not known to what extent these changes have been adopted by the dairy and feed industries in Canada.

Insufficient supply of nutrients and energy will limit productivity and jeopardize health (NRC 2001). Excess energy intake can result in overconditioning, which is not only costly, but can result in dystocia and reduced DM intake in early lactation (NRC 2001). Overfeeding of CP is also costly, and can result in high urea concentrations in blood, urine, and milk, reduced fertility (Canfield et al. 1990), increase energy expenditure for urea excretion (Fox et al. 1992) and contribute to environmental pollution (Baker et al. 1995). The right balance between RDP and RUP is essential to optimize post ruminal amino acid supply and enhance the efficiency of dietary N utilization (Wright et al. 1998). A minimum amount of peNDF, i.e., fibre, which stimulates chewing and promotes salivary buffering, is required to prevent ruminal acidosis (Mertens 1997). However, excess peNDF can limit feed intake (Allen 2000). The amount of NDF, forage NDF, and peNDF required in a lactating dairy ration to prevent milk fat depression and ruminal acidosis has been under debate, particularly in barley-based diets (Beauchemin et al. 1991). Currently, no standard and validated methods to measure peNDF in feeds exist, which prevents the determination of peNDF requirements (NRC 2001). Also, interactions between peNDF requirements and other chemical and physical characteristics of feed exist that have not been sufficiently studied (NRC 2001). As a result, the current edition of the Nutrient Requirements of Dairy Cattle (NRC 2001) does not provide requirements for dietary peNDF.

For all macro minerals considered essential, including Ca, P, K, Mg and Na, detrimental effects on animal performance can be demonstrated from feeding excessive amounts (NRC 2001). However, the dietary amounts required for optimal performance is well below amounts found to be detrimental to performance, and toxicity is rare (NRC 2001). High dietary inclusion rates of potassium in diets of prepartum cows are associated with high incidences of milk fever (Goff and Horst 1997). Of all dietary essential mineral elements, P represents the greatest potential risk if excess is released into the environment contaminating surface water and causing eutrophication (NRC 2001).

This survey was conducted to document nutritional practices and diet composition on Manitoba dairy farms, and study relationships between diet composition and milk production. This information will be used in the development of dairy research and teaching programs at the University of Manitoba and extension programs of the Manitoba Ministry of Agriculture and Food.

MATERIALS AND METHODS

Participating Farms

It was intended to include a representative sample of Manitoba dairy herds in the survey. A total of 75 dairy producers distributed across Manitoba out of 326 dairy producers participating in Western Canada Dairy Herd Improvement Services (WCDHIS) and 579 registered dairy producers were randomly selected and invited to participate

in the survey. Only milk producers that participated in WCDHIS were included in the survey, in order to assure that milk production data collected in a standardized fashion were available. Forty producers agreed to participate. The survey included 12.3% of milk producers participating in WCDHIS and 6.9% of all milk producers in Manitoba. Of these participants, 18 were located in Eastern Manitoba, 5 in the Interlake, 12 in Central Manitoba and 5 in Western Manitoba. The proportions of milk producers in each region of Manitoba that were invited to participate reflect the proportion of dairy farms located in that region. Thirty-six herds consisted of Holsteins, three of Jerseys and one of Brown Swiss. Participating producers were visited during October and November 2002. During these visits representative samples from all feeds, diets, and supplements, and the bulk milk tank were collected. The survey was approved by the Fort Garry Campus Animal Care Protocol Management and Review Committee and by the Research Ethics Board of the University of Manitoba. Each participating dairy producer agreed to participate in the survey by signing a consent form.

Feed and Milk Analyses

The DM contents of all feed samples were determined by drying at 60°C for 48 h. All dried feed and diet samples were analyzed for CP using the $\text{CuSO}_4/\text{TiO}_2$ Mixed Catalyst Kjeldahl procedure [Association of Official Analytical Chemists (AOAC) 1990, 988.05], RUP by incubation with the protease from *Streptomyces griseus* (Licitra et al. 1999), NDF (Van Soest et al. 1991) using α amylase (Sigma no. A3306; Sigma Chemical Co., St. Louis, MO), sodium sulphite and corrected for ash concentration adapted for Ankom200 Fibre Analyzer (Ankom Technology, Fairport, NY), ADF (AOAC 1990, 973.18), acid detergent lignin (ADL, AOAC, 1990), ether extract (AOAC 1990, 920.39), and ash (AOAC 1990, 942.05). Acid detergent fibre insoluble protein (ADFIP) and neutral detergent fibre insoluble protein (NDFIP) were determined by determining the CP fraction in the ADF and NDF, respectively. Feed, diet and mineral supplement samples were analyzed for Ca, P, K, Mg, and Na by inductively coupled plasma emission spectroscopy (AOAC 1990, 968.08) using an Atom Scan 25 plasma spectrometer (Thermo Jarrell Ash Corp, Grand Junction, CO) after acid digestion using a Perkin Elmer Optima 3000 spectrophotometer.

Particle size distributions were determined for all TMR, silage and chopped hay samples using the PSPS (Heinrichs 1996; Lammers et al. 1996). The PSPS used was equipped with two screens and a bottom pan. The diameters of holes of the screens were 19 and 8 mm for the top and middle screen, respectively. Approximately 150 g wet sample was placed on the top screen of the PSPS. The PSPS was shaken for a total of 40 times (five times in each direction, twice) (Heinrichs 1996). The contents of each fraction were weighed and subsequently dried and analyzed for DM as described earlier. Physically effective NDF (peNDF) was determined as the proportion of the dietary DM retained by the 8- and 19-mm screens of the PSPS multiplied by dietary NDF percentage.

For TMR-fed herds, the analysis of the TMR was used to determine the diet composition. For component-fed herds, the composition of the diet was estimated from the analyses of the feed ingredients and the inclusion rates of these ingredients as reported by the producer. Balances of NEL, metabolizable protein (MP), RDP, RUP, Ca, P, K, Mg, and Na were determined using NRC (2001). Herd average animal description and production data for each farm were obtained from CDHIS. Milk fat and protein percentages were obtained by analysis of the bulk milk tank samples. The inclusion rates of individual feeds were those reported by the producer. For the calculation of nutrient balances of component-fed herds average dietary inclusion rates of concentrates and forages were used. As no accurate data on feed intake could be obtained from the producers, dry matter intake (DMI) was estimated using the equation provided by NRC (2001). Both for TMR and component-fed herds, feeds that closely resembled the ingredients included in the diets were selected from the feed library of NRC (2001) and feed compositions were updated with the analyses of DM, CP, NDF, ADF, NDFIP, ADFIP, crude fat, ash, Ca, P, Mg, K, and Na that were conducted.

Samples from the bulk milk tank were collected in 50-mL containers and preserved with 2-bromo-2-nitropropane-1,3 diol. Milk samples were stored at 4°C until analyzed for fat and protein at the laboratory of the Manitoba Milk Producers (Winnipeg, MB) by near infrared analysis using the Milk-O-Scan 303AB (Foss Electric, Hillerød, Denmark). Milk urea nitrogen was analyzed in the laboratory of the Ontario Dairy Herd Improvement (Guelph, ON) by chemical analysis (Eurochem CL 10, Italy).

Statistical Analysis

Standard deviations, minimums, 25th percentiles, medians, 75th percentiles and maximums of measured variables were determined with the Means procedure of the SAS Institute, Inc. (1990). Pearson correlation coefficients between variables were determined with the Correlation procedure of the SAS Institute, Inc. (1990). Variables with a high correlation coefficient ($r > 0.75$) were not placed together in subsequent models due to collinearity. Linear regression models were developed using the Mixed procedure of the SAS Institute, Inc. (1990) to test the relationship between herd average milk yield, and the milk fat, milk protein and MUN content of the bulk milk tank (dependent variables) and variables describing the herd feeding management, and nutrient composition of the diet (independent variables). Diet description variables included DM, RDP, RUP, NDF, non-fibre carbohydrates (NFC), and peNDF. All diet composition variables were expressed as a percentage of the total DM, with the exception of DM, which was expressed as a percentage of the diet as is. Herd and feeding management variable were feeding practice (TMR or component-fed), breed (Holstein or other), average DIM and average age of the herd. These variables were included as fixed effects. Independent variables with a significance level greater than 0.25 were stepwise removed from the models.

RESULTS AND DISCUSSION

Average production data for the participating herds are given in Table 1. Average 305-d production for all cows

enrolled in WCDHIS in Manitoba in 2002 was 9033 kg (29.6 kg d⁻¹) milk, 325 (3.59%) kg milk fat, and 292 (3.23%) kg milk protein (WCDHIS 2003). This average milk yield differed only by 0.3 kg d⁻¹, which equalled 0.08 of the SD, from the median daily milk yield of the farms included in the survey. Average milk fat percentage differed by 0.22 percentage points, which equalled 0.69 of the SD, from the median milk fat percentage of the participating farms. Average milk protein percentage differed by 0.13 percentage points, which equalled 0.65 of the SD, from the median milk protein percentage of the farms included in the survey. Average herd size (lactating and non-lactating) in Manitoba in 2002 was 80 cows (WCDHIS, 2003), which was very close to the median size (lactating and non-lactating cows) of 79 of the herds included in the survey. Hence, it is assumed that the survey included a representative sample of Manitoba dairy farms.

More than 50% of participating farms had a bulk tank MUN greater than 15.6 mg dL⁻¹ (Table 1). Junker et al. (1998) recommended MUN levels between 10 and 16 MUN mg dL⁻¹ in order to limit excess N excretions. Due to differences in milk yield among cows, the MUN level of the bulk tank is not equal to the average of the MUN level of the individual cows in the herd. Nevertheless, the data suggest excess MUN resulting from excess dietary protein on nearly half of the participating farms. Decreased fertility has been associated with MUN concentrations greater than 19 mg dL⁻¹ (Butler et al. 1995). The MUN were higher than 19 mg dL⁻¹ only on three participating farms. These included two Jersey and one Holstein herds. Hence, in most herds with high MUN, this was not high enough to reduce reproductive performance. Average calving interval on these three herds was 15.1 mo, which was numerically higher than the survey average of 14.4 mo. However, as calving intervals are affected by many factors other than MUN and the calving intervals reported to WCDHIS were not considered to be very accurate, no conclusions on the basis of this numerical difference were drawn. Ogden et al. (2001b) observed that average herd MUN levels had a positive relationship with dietary CP, RDP, RUP, and feed costs, and a negative relationship with income over feed costs, forage to concentrate ratio (F:C), and non-fibre carbohydrates (NFC). In our study, the high dietary levels of CP and RDP are, in part, explained by the high inclusion rates of high protein forages such as alfalfa silage and alfalfa hay, and were not exclusively due to dietary inclusion rates of protein supplements. As a result, expected benefits of reducing the dietary CP and RDP levels are predominantly the reduction of N excretion to the environment.

Of the 40 participating producers, 25 (62.5%) fed TMR, whereas 15 (37.5%) practiced component feeding. Most TMR-fed herds (19 out of 25) used only one TMR for the entire herd. Five producers used separate TMR for low- and high-producing cows and one producer used separate TMR for fresh, high and medium/low-producing cows. The forage to concentrate ratio (DM basis) ranged from 38.1:61.9 to 67.4:32.6 among diets. The diets used by 20 (50%) of producers contained both silages and hay. On 15 farms (37.5%), the forage component of the diets consisted only of

Table 1. Milk production data and herd size on 40 representative dairy farms in Manitoba

Parameter	Median ^z	SD ^y	Minimum	25 perc ^x	75 perc ^w	Maximum
Milk yield (kg d ⁻¹)	29.3	3.97	18.4	27.1	31.3	41.5
Milk fat (%)	3.81	0.32	3.13	3.55	3.97	4.52
Milk protein (%)	3.36	0.20	3.07	3.25	3.45	4.00
MUN ^v (mg dL ⁻¹)	15.6	2.43	10.1	13.9	17.3	21.3
Lactating cows (no.)	65	38.6	35	51	95	212
Non-lactating cows (no.)	14	11.0	3	10	21	48
Average days in milk	171	23.8	131	152	189	227

^zMedian = 50% of farms below this value.^ySD = standard deviation.^x25 perc = 25% of farms below this value.^w75 perc = 75% of farms below this value.^vMUN = milk urea nitrogen.

silages. The forages used on the farms and the inclusion rates of these forages are given in Table 2. This table shows that feeding practices varied considerably among participating farms. Alfalfa and mixed alfalfa-grass silages and hay-lages were the most commonly used forages (75% of diets), followed by corn silage (58% of diets), and alfalfa and mixed alfalfa-grass hay (55% of diets). The dietary inclusion rates of these forages varied substantially among diets from lower than 10 to over 50% of DM (Table 2). More farms used ensiled forage than hay, but on half of the farms a combination of silage and hay was used.

Table 3 gives the description of lactating cow rations on participating dairy farms. A very large variation in dietary DM content was observed among farms. In the low DM diets the forage consisted of silage, whereas in the high DM diets the forage consisted of hay. Lahr et al. (1983) varied DM content in diets that otherwise did not differ in composition from 40 to 78% and found that DMI increased from 19.4 to 22.3 kg d⁻¹. Holter and Urban (1992) suggested that in cows fed diets ranging between 30 and 70% DM, DM content should not affect DMI. Moisture content has been reported to be negatively related to DMI, but this could be due to fermentation products rather than excess moisture (Allen 2000). NRC (2001) concludes that reports on the relationships between dietary DM content and DMI are conflicting, and that, therefore, no optimum DM content of the diet for maximum DMI is apparent.

NRC (2001) derived regression equations to determine the effects of CP, RDP, and RUP on milk yield and milk protein yield. Maximum milk yield and milk protein yield were obtained at 22 to 23% DM CP, but the marginal response of milk production due to increases in CP was low at the higher levels of CP. NRC (2001) warns that, although milk production may be increased by feeding diets with extremely high CP, the economic and environmental costs must be compared with lower CP diets. Maximum milk yield and milk protein yield occurred at 12.2% DM RDP, whereas milk protein yield increased linearly with increasing dietary RUP. Compared to the data used by NRC (2001) to derive these equations, CP and RDP were high and RUP was low. In 50% of herds RDP was greater than 12.2% DM. This suggests that on these farms more RDP was supplied than required for maximum milk yield and milk protein yield. NRC (2001) recommends that the adequacy of the protein

supply is evaluated on the basis of the balance of metabolizable protein and not on the basis of dietary concentrations of CP, RDP, and RUP. This balance is given in Table 4.

NRC (2001) recommends that the minimal dietary NDF inclusion rate should be between 25 and 33% DM and that the maximum dietary NFC inclusion rate should be between 36 and 44% DM. Beauchemin (1991) recommended that, due to the higher NDF content of barley grain compared to corn grain, diets based on barley should contain a minimum of 34% DM NDF. Most producers purchased their energy concentrates from feed companies, and the ingredient composition was, in most cases, not available. It is, however, assumed that most of these commercial energy concentrates contained barley grain. Table 3 shows that dietary NDF was below the minimum level recommended by Beauchemin (1991) on nearly half of the participating farms, which suggests the diets used on these farms might be limiting in forage NDF. Diets did not exceed the maximum NFC content recommended by NRC (2001). NRC (2001) concluded that in most situations, dietary fat should not exceed 6–7% of dietary DM. Table 3 shows that fat inclusion rate did not exceed this maximum in any of the diets. Requirements of macrominerals are given by NRC (2001) in grams per day, and not in dietary concentrations. Hence, the adequacy of the macromineral supply was determined on the basis of the balances of these minerals (Table 3) and not on the basis of their dietary concentrations. The variability of peNDF among farms was substantial (Table 3). At the present time, no standard validated methods for the measurement of peNDF and for the establishment of peNDF requirements exist (NRC 2001). Until such methods are available, interpretation of this variability is difficult.

Balances of NEI, MP, RDP, RUP, Ca, P, K, Mg, and Na as determined by NRC (2001) are given in Table 4. This table shows that the supply of NEI closely matched the requirements. On average, diets contained 6.2% more NEI than required, which might be attributed to diets being formulated to match the nutrient requirements of cows producing more than the average of the group. The supply of MP also closely matched requirements. On average, diets contained 35.9% more RDP than required, which resulted in an average RDP balance of 758 g d⁻¹. As this balance will have been excreted, nitrogen excretions to the environment were higher than necessary. The balance of RUP was, on average

Table 2. Inclusion rates of forages in lactating cow rations on 40 representative dairy farms in Manitoba

Forage	Percentage of farms using (%)	Inclusion rates (% DM)				
		Median ^z	Minimum	25 perc ^y	75 perc ^x	Maximum
Corn silage	58	21.0	6.4	16.8	28.1	57.4
Alfalfa and mixed silages and haylages	75	28.9	9.0	19.8	39.2	53.4
Alfalfa and mixed hays	55	13.4	1.7	6.6	26.5	65.0
Barley silage	23	16.6	7.6	13.6	21.6	24.5
Oatlage	7.5	13.8	12.9	13.4	14.5	15.2
Grass hay	13	18.4	15.4	16.9	23.4	30.3
Grass silage ^w	5	—	10.5	—	—	23.3

^zMedian = 50% of farms below this value.^y25 perc = 25% of farms below this value.^x75 perc = 75% of farms below this value.^wMinimum and maximum reported only as only used in two diets.**Table 3. Description of lactating cow rations on 40 representative dairy farms in Manitoba**

	Unit	Median ^z	SD ^y	Minimum	25 perc ^x	75 perc ^w	Maximum
Dry mater	%	59.0	15.5	43.0	53.3	66.8	95.0
Crude protein	% DM	18.3	1.63	14.6	16.9	19.4	21.4
Rumen degradable protein	% DM	12.2	1.4	9.3	11.3	13.1	15.7
Rumen undegradable protein	% DM	5.7	1.3	4.0	5.1	6.7	8.8
Neutral detergent fibre	% DM	34.8	4.20	27.4	31.2	36.6	48.2
Acid detergent fibre	% DM	21.4	3.31	15.1	19.4	23.8	32.4
Acid detergent lignin	% DM	4.1	1.04	2.0	3.2	4.6	5.9
Non-fibre carbohydrates	% DM	34.7	4.54	24.1	31.8	38.6	44.5
Crude fat	% DM	3.4	1.02	2.3	2.8	4.0	6.4
Ash	% DM	8.6	1.39	4.1	7.9	9.4	11.2
Calcium	% DM	1.03	0.13	0.77	0.91	1.16	1.60
Phosphorus	% DM	0.48	0.07	0.33	0.42	0.54	0.69
Potassium	% DM	1.60	0.13	1.03	1.49	1.76	2.16
Magnesium	% DM	0.43	0.06	0.30	0.36	0.47	0.55
Sodium	% DM	0.28	0.04	0.09	0.22	0.36	0.55
Physical effective NDF	% DM	17.0	6.67	7.7	11.9	21.9	37.3

^zMedian = 50% of farms below this value.^ySD = standard deviation.^x25 perc = 25% of farms below this value.^w75 perc = 75% of farms below this value.

15.8% below requirement and this balance ranged from 42.1% less than requirements to 22.5% more than requirements. This suggests that milk production on many farms could be improved by increasing the dietary content of RUP.

Balances for macrominerals varied more among farms than the balances for NEL, MP, RDP, and RUP. On average, dietary supply of Ca, P, K, Mg, and Na exceeded requirements for these minerals by 74.6, 51.9, 44.9, 116.1, and 4.2%, respectively. These balances were calculated based on the average milk production on the farm and estimated DMI. As diets are generally formulated to match the nutrient requirements of cows producing more than the average of the group, excesses in the supply of nutrients can be expected. However, these excesses should not be as high as those observed from Ca, P, K, and Mg. None of the balances of these minerals was so high that toxicity could result (NRC 2001).

Nutrient balances were determined using DMI estimated by NRC (2001). Estimated DMI using the equation from NRC (2001) has a bias of -0.27 kg d^{-1} and a mean square prediction error of $3.31 \text{ kg}^2 \text{ d}^{-1}$, and does not account for diet composition, weather, feeding method, feeding frequency, and sequence of feeding (NRC 2001). Hence, esti-

mated DMI can vary substantially from actual DMI. However, it was believed that estimating DMI from information provided by the producer would have been less accurate than estimating DMI using the equation provided by NRC (2001). Despite this drawback in the estimation of nutrient balances, it is believed that even allowing for a safety margin, dietary inclusion rates of Ca, P, K, and Mg could have been reduced in order to reduce excretion of these minerals in the environment without negatively affecting productivity and fertility.

Results of particle size analysis of forages and TMR on Manitoba dairy farms using the PSPS are given in Table 5. Recommendations for particle size distributions are that between 50 and 60% of particles of silages and between 40 and 60% of TMR particles should be retained by the 8- and 19-mm screens of the PSPS (Heinrichs 1996). These recommendations have been developed on the assumption that feed particles retained by these screens are physically effective, i.e. contribute to rumen buffering. In just over 25% of TMR, less than the recommended 40% of feed particles were retained by the PSPS screens. Only 35% of alfalfa silage samples and 30% of the corn silage samples were within the recommended range. In 40% of corn silage sam-

Table 4. Balances of NEL, MP, RDP, RUP, calcium, phosphorous, potassium, magnesium, and sodium relative to estimated NRC (2001) requirements on 40 representative dairy farms in Manitoba

Item ^z	Unit	Median ^y	SD ^x	Minimum	25 perc ^w	75 perc ^v	Maximum
NEL	Mcal d ⁻¹	2.05	2.7	-1.65	1.2	2.65	7.7
NEL	% of requirement	6.2	4.7	-4.9	3.6	8.1	19.4
MP	g d ⁻¹	-142	159	-433	-210	-12	174
MP	% of requirement	-0.6	7.2	-18.8	-9.6	8.5	7.2
RDP	g d ⁻¹	758	379	161	569	1185	1597
RDP	% of requirement	35.9	17.3	7.5	25.4	54.4	65.9
RUP	g d ⁻¹	-183	206	-582	-266	-16	226
RUP	% of requirement	-15.8	17.4	-42.1	-22.5	-1.3	22.5
Calcium	g d ⁻¹	43.0	31.2	-19.0	28.5	56.3	115.0
Calcium	% of requirement	74.6	53.6	-34.6	50.2	98.9	215.1
Phosphorus ^u	g d ⁻¹	26.5	13.4	3.0	14.0	34.3	56.0
Phosphorus ^u	% of requirement	51.9	26.5	5.9	28.5	70.6	116.7
Potassium ^u	g d ⁻¹	98.5	55.8	-16.0	60.8	120.5	248.0
Potassium ^u	% of requirement	44.9	27.4	-8.0	29.2	62.4	120.4
Magnesium ^u	g d ⁻¹	7.4	3.0	2.1	6.3	9.6	14.5
Magnesium ^u	% of requirement	116.1	47.9	32.8	101.1	162.8	237.7
Sodium ^u	g d ⁻¹	4.2	23.6	-39.4	-7.3	16.5	75.0
Sodium ^u	% of requirement	9.7	54.9	-88.7	-16.4	34.6	174.4

^zNEL = net energy for lactation, MP = metabolizable protein, RDP = rumen degradable protein, RUP = rumen undegradable protein.

^yMedian = 50% of farms below this value.

^xSD = standard deviation.

^w25 perc = 25% of farms below this value.

^v75 perc = 75% of farms below this value.

^uValues are in total absorbable supplied.

ples between 60 and 70% of feed particles were retained by the PSPS screens, which indicates that these samples were too coarse. In 30% of corn silage samples, the proportion of particles retained by the PSPS screens was insufficient, indicating that these samples were too fine. Based on PSPS data, 30% of alfalfa silage samples were too coarse and 35% of alfalfa silage samples were too fine. Deviating from particle size distribution of corn silage, alfalfa silage and TMR recommended by Heinrichs (1996), could affect dairy cow health and production, as peNDF can result in insufficient rumen buffering and ruminal acidosis (Mertens 1997) and excess peNDF can limit feed intake (Allen 2000). The optimal range of dietary peNDF is expected to depend on forage source and concentrate source of the diet, as these factors affect both the requirement for rumen buffering and the capacity for rumen buffering of the diet (NRC 2001; Calberry et al. 2003; Plaizier 2004).

Of all the herds, feeding, and dietary variables considered, only breed significantly affected milk yield and milk fat yield; breed, DM, and RUP affected milk protein; and DIM, RDP, RUP, and NDF affected MUN (Table 6). Breed effects on milk yield and milk composition were expected, as Jerseys and Brown Swiss produce less milk volume, but have higher milk fat and milk protein contents than Holsteins (WCDHIS 2003). The positive relationship between RUP and milk protein content was expected, as RUP was below requirements on most farms (Table 3). The RDP was in excess of requirements on all farms, but these excesses must not have affected milk production. The energy required for the excretion of urea due to excess RDP could lower the NEL available for milk production and, thereby, reduce milk production (NRC 2001). This could, however, not be demonstrated in our study. Differences in

dietary contents of NFC and NDF among farms could result in differences in dietary NEL among farms. That these differences in dietary NFC and NDF contents did not affect milk production, might be explained by the good match between NEL requirements and NEL supply on most farms.

A negative relationship between dietary DM and milk protein was observed. Dietary DM varied considerably among farms (Table 3). This variation in DM was caused by differences in the forage to concentrate ratio, DM of silages, and inclusion of hay vs. silage. Lahr et al. (1983) and Beauchemin et al. (2003) did not find that substituting alfalfa hay for alfalfa silage affected milk protein. If the ratio between hay and silage affects milk protein, then this might be due to differences between these forages other than DM, such as fibre content and digestibility. It is possible that hay was generally harvested at a more mature stage than silage, which would make hay-based diets less digestible than silage-based diets. Nevertheless, the reason for the effect of DM on milk protein remains unclear.

Dietary peNDF did not affect milk production. Due to the large range in dietary peNDF among farms and the large proportion of alfalfa silage, corn silage, and TMR samples that were finer than recommended by Heinrichs (1995) (Table 5), an effect of peNDF on milk production, and especially on milk fat, could be expected. Using data from previous studies, Mertens (1997) observed a curvilinear relationship between physical effective NDF and milk fat, and found that a reduction in dietary physical effective NDF content caused a large reduction milk fat percentage in fine diets compared to coarse diets. In controlled studies, Kononoff et al. (2000), Krause et al. (2002), and Beauchemin et al. (2003) did not observe an effect of dietary particle size on milk yield and milk composition, whereas Yang et al. (2001) and Calberry et al. (2003) found that

Table 5. Results of particle size analysis of forages and total mixed rations (TMR) on Manitoba dairy farms using the Penn State Particle Separator (PSPS)

	Median ^z	SD ^y	Min. ^x	25 perc.	75 perc.	Max. ^w	N ^v	Recc. ^u
Alfalfa silage								
Top screen (>19 mm) (%)	18.4	11.2	5.6	11.4	24.4	69.6	25	
Bottom screen (>8 mm) (%)	43.6	9.3	25.7	38.5	52.0	47.5		5–10
Bottom pan (%)	34.4	13.8	11.7	28.3	43.1	63.0		40–50
peNDF (% DM)	27.1	8.1	15.1	23.6	32.2	49.0		40–50
Corn silage								
Top screen (>19 mm) (%)	5.5	3.4	2.9	5.5	8.7	18.9	24	
Bottom screen (>8 mm) (%)	40.2	12.5	18.4	40.2	59.0	70.5		5–10
Bottom pan (%)	33.9	12.2	21.9	33.9	51.7	72.5		40–50
peNDF (% DM)	21.6	5.6	13.4	19.6	28.5	33.5		40–50
TMR								
Top screen (>19 mm) (%)	11.9	10.8	2.6	7.6	18.6	68.8	30	
Bottom screen (>8 mm) (%)	30.8	8.5	18.4	26.4	39.7	49.5		5–10
Bottom pan (%)	52.8	12.6	22.0	46.0	61.6	72.5		30–50
peNDF (%DM)	16.1	6.50	7.7	11.2	19.6	37.3		40–60

^zMedian = 50 % of farms below this value.^ySD = standard deviation.^xMin. = minimum.^wMax. = maximum.^vN = number of samples.^uRecommended. Source = Heinrichs (1996).**Table 6. Parameter estimates of equations describing the relationships between dependent milk production variables (milk yield, milk fat percentage, milk protein percentage, and milk urea nitrogen (MUN)) and independent herd and feeding variables (breed, TMR vs. component feeding, average age, average days in milk (DIM), and diet composition). Only estimates of significant independent variables ($P < 0.05$) are given**

Dependent variables	Independent variables	Value	Estimate	SE of estimate	P value
Milk yield (kg d ⁻¹)	Intercept	—	13.98	4.69	0.0051
	Breed	Holstein	7.54	2.39	0.0032
		Jersey and Brown Swiss	referent	—	—
Milk fat (%)	Intercept	—	3.15	0.46	<0.001
	Breed	Holstein	-0.38	0.15	0.02
		Jersey and Brown Swiss	referent	—	—
Milk protein (%)	Intercept	—	2.98	0.39	<0.001
	Breed	Holstein	-0.34	0.11	0.003
		Jersey and Brown Swiss	referent	—	—
	DM (%)	—	-0.005	0.002	0.014
	RUP (% DM)	—	0.07	0.02	0.007
MUN (mg dL ⁻¹)	Intercept	—	-44.07	4.76	0.076
	DIM (d)	—	0.03	0.30	0.03
	RUP (% DM)	—	1.97	0.30	<0.0001
	RDP (% DM)	—	0.84	0.25	0.002
	NDF (% DM)	—	0.31	0.15	0.04

increasing dietary particle size numerically increased milk fat concentration, but did not affect milk yield and milk protein percentage. A reduction in dietary particle size could reduce milk fat percentage when it is accompanied by a reduction in rumen pH, as a reduction in rumen pH result in an increase in *trans* fatty acids and a reduction in de novo milk fat synthesis (Griinari et al. 1998). Rumen pH is, however, only affected by peNDF below a peNDF threshold (Mertens 1997). Also, effects of dietary particle size on milk yield and milk composition might be obscured due to many other factors, such as rumen buffering capacity of forage (McBurney et al. 1983) and the inclusion rate of inorganic buffers (NRC 2001) that also affect rumen pH. This is illustrated by Beauchemin et al. (2003), who observed that

increased intake of peNDF, estimated as the proportion of DM retained by the 8- and 19-mm PSPS sieves multiplied by the dietary NDF content, numerically increased rumination time, eating salivary output, and mean rumen pH, significantly decreased duration of rumen pH below 5.8, but did not significantly affect milk fat percentage and milk yield. Yang et al. (2001) used the same calculation for peNDF as Beauchemin et al. (2003), and found no significant relationships between various measures of peNDF, milk fat content, and rumen pH.

Calberry et al. (2003) and Leonardi and Armetano (2003) demonstrated that cows selected against coarse feed particles in favour of fine feed particles. Hence, the particle size distribution and peNDF of ingested feed particles can be

finer than that of the TMR, and the chewing, saliva production, and rumen buffering can be less than what could be expected from the particle size distribution of the TMR.

The effect of peNDF on milk production remains inconclusive. The lack of a relationship between peNDF and milk production in our study could be due to the majority of diets having a peNDF higher than the threshold below which further reduction of peNDF affects milk production. Also, in our study peNDF was calculated as the proportion of DM retained by the 8- and 19-mm screens of the PSPS multiplied by dietary NDF. Other methods for the estimation of peNDF exist (Mertens 1997; Yang et al. 2001; Beauchemin et al. 2003; Calberry et al. 2003). The method used in this study is easier to conduct on farm than these other methods, but it might not be the most accurate measure of chewing, saliva production and rumen buffering (Yang et al. 2001; Beauchemin et al. 2003). In order to determine the relationship between peNDF and milk production, regression equations between the various measures of peNDF, milk yield, milk fat yield, and milk protein yield of cows fed fine diets containing various combinations of forages and concentrates need to be derived.

Similar to our study, Godden et al. (2001b) also found that MUN was positively related to RDP and RUP. Such a relationship is expected, as high RDP and RUP will increase the levels of both blood urea nitrogen and MUN (Van Soest 1994). In contrast to Godden et al. (2001b) a negative relationship between MUN and dietary NFC could not be demonstrated. This might be due to the relatively small size of the survey compared to the study from Godden et al. (2001b) and that in the latter study farms were monitored for a prolonged period, whereas the farms included in our study were only visited once. Godden et al. (2001b) also did not find an effect of feeding management (TMR or component feeding) on MUN.

Average DIM correlated positively with MUN (Table 6). This agrees with Godden et al. (2001a), who observed that MUN was lower during the first 60 d of lactation, higher between 60 and 150 DIM, and lower after approximately 150 DIM. These authors explained this effect of state of lactation on MUN by changes in nutrient composition or feeding programs among different stages of lactation. Godden et al. (2001b), however, did not find that stage of lactation affected MUN. Dietary NDF correlated positively with MUN also. As NDF and NFC were strongly negatively correlated. This agrees with Godden et al. (2001b), who observed a negative correlation between MUN and NFC, indicating a higher efficiency of N utilization in high concentrate diets compared to low concentrate diets.

CONCLUSIONS

Feeding practices varied considerably among farms. On 62.5% of farms TMR feeding was used, whereas 37.5% of farms used component feeding. Only 24% of TMR-fed herds had more than one TMR, which is a limitation to meeting the nutrient requirements of individual cows. The supply of NEI closely matched NRC (2001) requirements, but RDP, Ca, P, Mg, and K were in excess of NRC (2001) requirements on most farms. Reduction in the dietary con-

tents of these nutrients will reduce nutrient excretions to the environment without reducing milk production. The RUP was below (NRC 2001) requirements on nearly all farms, and dietary RUP content was positively related to milk protein content. This shows that increasing dietary RUP content could improve milk protein production on Manitoba dairy farms. Measurement of MUN proved to be a useful tool to monitor protein nutrition and the efficiency of N utilization. The peNDF, determined as proportion of the dietary DM retained by the 8- and 19-mm screens of the PSPS multiplied by dietary NDF was not related to milk yield and milk composition. This could be due to the many dietary, genetic, and management factors that affect rumen pH and milk fat content. More controlled studies on the effects of dietary particle size and the interactions between dietary particle size, forage source, concentrate source, and forage to concentrate ratio on rumen pH, milk yield and milk composition are required before recommendations for peNDF can be finalized.

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Allen, M. S. 2000. Effects of diet on short-term regulation of feed intake by lactating dairy cattle. *J. Dairy Sci.* **83**: 1598–1624.

Association of Official Analytical Chemists. 1990. Official methods of analysis. 15th ed. AOAC, Arlington, VA.

Baker, L. D., Ferguson, J. D. and Chalupa, W. 1995. Responses in urea and true protein of milk to different protein feeding schemes for dairy cows. *J. Dairy Sci.* **78**: 2424–2434.

Beauchemin, K. A. 1991. Effects of dietary neutral detergent fiber concentration and alfalfa hay quality on chewing, rumen function, and milk production of dairy cows. *J. Dairy Sci.* **74**: 3140–3151.

Beauchemin, K. A., Farr, B. I. and Rode, L. M. 1991. Enhancement of the effective fiber content of barley-based concentrates fed to dairy cows. *J. Dairy Sci.* **74**: 3128–3139.

Beauchemin, K. A., Yang, W. Z. and Rode, L. M. 2003. Effects of particle size of alfalfa based dairy cow diets on chewing activity, ruminal fermentation, and milk production. *J. Dairy Sci.* **86**: 630–643.

Butler, W. R., Calaman, J. J. and S.W. Beam, S.W. 1995. Plasma and milk urea nitrogen in relation to pregnancy in dairy cattle. *J. Anim. Sci.* **74**: 858–865.

Calberry J. M., Plaizier, J. C., Einarson, M. S. and McBride, B.W. 2003. Replacing chopped alfalfa hay with alfalfa silage in a high barley concentrate total mixed ration. *J. Dairy Sci.* **86**: 3611–3619.

Canfield R. W., Sniffen, C. J. and Butler, W. R., 1990. Effects of excess degradable protein on post-partum reproduction and energy balance in dairy cattle. *J. Dairy Sci.* **73**: 2342–2349.

- Fox, D. G., Sniffen, C. J., O'Connor, J. D., Russel, J. B. and Van Soest, P. J. 1992.** A net carbohydrate and protein system for evaluating cattle diets: III. Cattle requirements and diet adequacy. *J. Anim. Sci.* **70**: 3578–3596.
- Godden, S. M., Lissemore, K. D., Kelton, D. F., Leslie, K. E. and Lumsden, J. H. 2001a.** Factors associated with milk urea concentrations in Ontario dairy cows. *J. Dairy Sci.* **84**: 107–114.
- Godden, S. M., Lissemore, K. D., Kelton, D. F., Leslie, K. E. and Lumsden, J. H. 2001b.** Relationships between milk urea concentrations and nutritional management, production, and economic variables in Ontario dairy herds. *J. Dairy Sci.* **84**: 1128–1139.
- Goff, J. P. and Horst, R. L. 1997.** Effects of adding potassium or sodium, but not calcium to prepartum rations on milk fever in dairy cows. *J. Dairy Sci.* **80**: 176–186.
- Griinari, J. M., Dwyer, D. A., McGuire, M. A., Bauman, D. E., Palmquist D. L. and Heinrichs, J. 1996.** Evaluating particle size of forages and TMRs using the Penn State Particle Size Separator. Pennsylvania, PA. pp. 1–9.
- Holter, J. B. and Urban, W. E., Jr. 1992.** Water partitioning and intake prediction in dry and lactating Holstein cows. *J. Dairy Sci.* **75**: 1472–1479.
- Jonker, J. S., Kohn, R. A. and Erdman, R. A. 1998.** Using milk urea nitrogen to predict nitrogen excretion and utilization efficiency in lactating cows. *J. Dairy Sci.* **81**: 2681–2692.
- Kononoff, P. J., Mustafa, A. F., Christensen, D. A. and McKinnon, J. J. 2000.** Effects of barley silage particle length and effective fiber on yield and composition of milk from dairy cows. *Can. J. Anim. Sci.* **80**: 749–752.
- Krause, K. M., Combs, D. K. and Beauchemin, K. A., 2002.** Effects of forage particle size and grain fermentability in midlactation cows. I. Milk production and diet digestibility. *J. Dairy Sci.* **85**: 1936–1946.
- Lahr, D. A., Otterby, D. E., Johnson, D. G., Linn, J. G. and Lundquist, R. G. 1983.** Effects of moisture content of complete diets on feed intake and milk production by cows. *J. Dairy Sci.* **66**: 1891–1900.
- Lammers, B. P., Buckmaster, D. R. and Heinrichs, A. J. 1996.** A simple method for the analysis of particle sizes of forages and total mixed rations. *J. Dairy Sci.* **79**: 922–928.
- Leonardo, C. and Armetano, E. 2003.** Effect of quantity, quality and length of alfalfa hay on selective consumption by dairy cows. *J. Dairy Sci.* **86**: 557–556.
- Licitra, G., Van Soest, P. J., Schadt, I., Carpino S. and Sniffen, C. J. 1999.** Influence of the concentration of the protease from *Streptomyces griseus* relative to ruminal protein degradability. *Anim. Feed Sci. Technol.* **77**: 99–113.
- McBurney, M. I., Van Soest, P. J. and Chase, L. E. 1983.** Cation exchange capacity and buffering capacity of neutral detergent fibres. *J. Sci. Food Agric.* **34**: 910–916.
- Mertens, D. R. 1997.** Creating a system for meeting the fiber requirements of dairy cows. *J. Dairy Sci.* **80**: 1463–1481.
- National Research Council. 1989.** Nutrient requirements of dairy cattle. 6th rev. ed. National Academy Press, Washington, DC.
- National Research Council. 2001.** Nutrient requirements of dairy cattle. 7th rev. ed. National Academy Press, Washington, DC.
- Plaizier, J.C. 2004.** Replacing chopped alfalfa hay with alfalfa silage in barley brain and alfalfa based total mixed rations for lactating dairy cows. *J. Dairy Sci.* **87**: 2495–2505.
- SAS Institute, Inc. 1990.** SAS user's guide. Statistics. Version 6 ed. 1990. SAS Institute, Inc., Cary, NC.
- Van Soest, P. J., Robertson, J. B. and Lewis, B. A. 1991.** Methods for dietary fiber, neutral detergent fibre, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* **74**: 3583–3597.
- Van Soest, P. J. 1994.** Nutritional ecology of the ruminant 2nd ed. Cornell University Press, Ithaca, NY.
- Western Canada Dairy Herd Improvement Services. 2003.** WCDHIS herd improvement report. Western Canada Dairy Herd Improvement Services Edmonton, AB.
- Wright, T. C., Moscardini, S., Luimes, P. H., Susmel, P. and McBride, B. W. 1998.** Effects of rumen-undegradable protein and feed intake on nitrogen balance and milk protein production in dairy cows. *J. Dairy Sci.* **81**: 784–793.
- Yang, W. Z., Beauchemin, K.A. and Rode, L.M. 2001.** Effects of grain processing forage to concentrate ratio, and forage particle size on rumen pH and digestion by dairy cows. *J. Dairy Sci.* **84**: 2203–2216.

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